

WILLMAR, MN, USA

WMO: 720386

Lat: 45.117N

Lon: 95.133W

Elev: 1129

StdP: 14.11

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 00125

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-16.5	-10.5	-22.2	1.7	-14.4	-17.0	2.3	-10.0	32.3	13.4	28.3	14.3	6.2	300	0.585

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.0	89.8	72.8	86.2	71.8	83.6	70.5	76.8	85.4	74.8	82.7	72.9	80.7	11.1	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
73.3	129.2	81.8	72.3	124.5	80.7	70.3	116.2	78.5	41.0	85.9	39.0	82.1	37.3	81.0	84.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.7	23.6	20.1	-21.4	94.9	6.6	2.1	-26.2	96.4	-30.0	97.6	-33.7	98.8	-38.5	100.4
			-21.8	80.2	6.3	2.4	-26.3	81.9	-30.1	83.3	-33.6	84.6	-38.2	86.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	44.6	12.6	17.2	30.3	44.9	57.5	68.0	71.7	69.0	62.1	48.1	32.7	19.0	(6)
	DBStd	23.03	13.72	13.23	12.86	10.00	8.80	6.71	5.54	5.44	8.46	9.42	11.35	11.73	(7)
	HDD50	4581	1160	919	619	210	30	0	0	0	8	149	526	960	(8)
	HDD65	8066	1625	1339	1076	605	266	45	11	23	154	529	968	1425	(9)
	CDD50	2597	0	0	8	57	261	539	674	589	371	90	8	0	(10)
	CDD65	605	0	0	0	2	32	133	219	147	67	5	0	0	(11)
	CDH74	5365	0	0	1	42	402	1192	1939	1134	582	72	1	0	(12)
	CDH80	1565	0	0	0	11	137	371	591	276	163	16	0	0	(13)
	WSAvg	8.9	9.5	9.4	10.0	10.8	10.1	8.5	6.6	6.2	7.6	9.2	9.5	9.1	(14)
	PrecAvg	29.7	0.8	0.8	1.6	2.9	3.2	4.9	3.8	3.9	3.0	2.5	1.5	0.8	(15)
(16) Precipitation	PrecMax	38.3	1.9	1.9	3.3	6.9	7.0	8.6	6.7	7.5	8.4	6.9	4.3	2.0	(16)
	PrecMin	21.5	0.1	0.3	0.5	0.8	1.0	2.5	0.8	0.5	0.6	0.6	0.0	0.2	(17)
	PrecStd	4.3	0.6	0.5	0.8	1.5	1.5	1.9	1.5	1.8	1.9	1.6	1.3	0.5	(18)
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	43.4	52.2	67.8	79.3	90.4	92.3	91.5	90.2	90.0	81.8	66.2	47.7	(19)
		MCWB	37.5	48.3	57.3	61.1	65.3	72.3	76.7	74.8	70.9	63.4	56.5	42.3	(20)
	2%	DB	36.7	43.3	60.5	72.5	83.6	88.2	89.7	85.9	84.2	73.4	60.5	41.3	(21)
		MCWB	34.2	39.6	53.4	55.9	63.8	70.3	76.0	74.2	69.9	60.3	50.9	37.3	(22)
	5%	DB	34.2	37.5	53.7	66.2	78.7	83.9	85.9	82.3	81.1	68.2	54.0	36.9	(23)
		MCWB	32.6	35.0	48.9	53.1	61.5	68.6	73.8	71.2	67.6	57.1	47.4	34.6	(24)
	10%	DB	30.5	35.9	46.2	61.5	72.8	81.1	82.4	80.9	76.7	63.3	47.8	34.4	(25)
		MCWB	29.6	34.0	42.5	50.3	59.2	66.8	71.5	69.9	65.4	54.4	43.1	32.8	(26)
	0.4%	WB	38.2	47.8	60.1	63.2	71.3	76.9	80.6	78.3	75.5	67.5	58.5	44.0	(27)
		MCDB	42.7	52.2	63.9	74.4	83.3	87.3	87.7	85.9	83.7	75.3	64.4	47.3	(28)
(30) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	34.8	39.4	53.6	58.3	66.9	74.2	77.7	75.5	72.6	62.7	52.1	37.8	(29)
		MCDB	36.7	42.3	58.2	68.6	77.7	83.6	86.4	82.8	81.3	70.3	57.2	40.6	(30)
	5%	WB	32.5	36.2	47.6	54.7	64.0	71.3	75.4	73.5	70.3	59.0	47.2	35.1	(31)
		MCDB	34.0	38.3	51.8	63.9	73.5	79.9	83.2	80.5	78.1	66.7	52.7	36.9	(32)
	10%	WB	29.9	33.5	42.5	51.8	61.2	68.7	73.5	71.4	67.4	55.1	43.0	32.8	(33)
		MCDB	30.8	35.5	47.0	60.9	69.9	77.7	81.3	78.4	74.6	62.4	47.7	34.5	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	16.2	17.4	16.4	19.6	20.5	19.1	19.0	19.3	21.3	19.4	16.2	15.0	(35)
		MCDBR	17.8	17.6	24.6	29.3	28.7	23.7	21.0	21.3	25.1	27.9	24.6	18.1	(36)
	5% WB	MCWBR	15.5	14.8	17.1	16.5	13.8	11.1	11.2	11.9	13.0	15.4	16.5	15.1	(37)
		MCDBR	16.6	16.5	23.5	24.7	22.8	19.7	19.1	19.2	21.3	23.2	22.8	17.0	(38)
		MCWBR	14.6	13.6	16.9	15.6	13.2	11.6	11.4	12.1	12.8	14.7	16.1	14.4	(39)
(40) Clear-Sky Solar Irradiance	taub		0.260	0.275	0.307	0.363	0.388	0.392	0.392	0.402	0.363	0.315	0.279	0.255	(40)
	taud		2.449	2.404	2.428	2.327	2.321	2.363	2.354	2.312	2.400	2.528	2.538	2.476	(41)
	Ebn at Noon		277	292	295	284	279	278	276	269	272	274	268	268	(42)
	Edh at Noon		22	28	31	38	39	38	38	38	32	24	20	19	(43)
(44) All-Sky Solar Radiation	RadAvg		562	901	1204	1491	1717	1910	2054	1703	1316	844	544	401	(44)
	RadStd		68	104	114	135	131	132	71	138	98	117	51	46	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (68 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page